



LXX

Co nám může říct měření tučňáků?



Vztahy dvou proměnných

cuni-natur-biostatistics.github.io/materialy.html

Co vidíme v měření tučňáků?

TODO: Ukažte jeden konkrétní obrázek, měření nebo jiný háček k úvodní otázce. Nechte studenty nejprve popsat, co vidí.

Výsledky učení

Po absolvování této přednášky budete schopni:

- TODO: výsledek učení 1
- TODO: výsledek učení 2
- TODO: výsledek učení 3

Obsah přednášky

Snímek 1

TODO: obsah snímku

```
1 # TODO: ukázka kódu v R
```

Shrnutí

- TODO: klíčové body přednášky

Slide Creator's Guide

Slides in this section are intended for lecture authors, not students. Delete this section and the `# Tools Overview` section before using as an actual lecture.

Golden Rules

Rule 1 — Data story first

WRONG

Pearson correlation

Pearson's correlation coefficient measures the strength of the linear relationship between two variables...
[formula]

RIGHT

How do we turn the shape of a cloud into a number?

[scatter plot the students have seen for several slides already]
What properties would such a number need to have?

→ **Correlation coefficient** the name arrives as the answer

A statistical term never opens a concept as a heading. Data first → observation → question → only then the name.

Rule 2 — One idea per slide

Every slide communicates one thing.

If you need to say two things, use two slides.

If you need to say three things, add an interaction between them.

One message. One slide. Contribute to understanding — not to overload.

Test: cover the heading — is the main idea of the slide still clear from the visual or a single sentence?

Rule 3 — Interaction every 10–15 minutes

SIX INTERACTION TYPES

Type	When to use
Prediction question	Before revealing a result, graph, or test output
Conceptual MCQ	To surface a common misconception; ideally with a polling tool
Think–Pair–Share	After a new concept; 30 s thinking + 1–2 min pair discussion
Silent write	Before discussion or reveal; each student writes one sentence
Interpretation task	After showing a graph, model output, or table

Rule 4 — Always frame biologically

Every statistical method must be grounded in a concrete biological question.

Four steps of the biological frame:

1. **Biological question** — “we want to find out whether...”
2. **Data** — description, type, source
3. **Statistical method** — what and why
4. **Interpretation in biological context** — what the result says about the world, not about numbers

Never start with “today we’ll learn the t-test”. Always start with “we want to know whether...”

Rule 5 — Alternate visual patterns (3-of-5 rule)

At most 3 of 5 consecutive slides may use a named component as the primary visual element.

A plain slide — a bold question on an empty background, a full-width graph, a single prominent number — carries more weight precisely because the surrounding slides are more structured.

3 of 5 — the law of slide visual contrast · review the block before finalising

Components: `.card-*`, `.panel-*`, `.blockquote`, `.takeaway-strip`. If you reach 4+ in a row, convert one slide to a plain visual or an unframed question.

Five Slide Families

Five slide families

Family	Goal	Typical component
Interaction	Prediction / MCQ / TPS	bold question, .panel-question
Data reveal	New data or output	full-width graph, plain slide
Interpretation	Read the result	.panel-result, .takeaway-strip
Rule /	Heuristic,	.card-

VARIETY OF FAMILIES

In one lecture, cycle through all five families. No block of 5+ slides should belong to only one family.

Tools Overview

`.box-*` aliases — when to use what

Alias	Maps to	Weight	Colour	When to use
<code>.box-question</code>	<code>.panel-question</code>	panel (lighter)	orange left border	MCQ, reflection question
<code>.box-result</code>	<code>.panel-result</code>	panel (lighter)	indigo left border	main conclusion, analysis result
<code>.box-data</code>	<code>.panel-data</code>	panel (lighter)	olive left border	dataset description, data context
<code>.box-rule</code>	<code>.card-rule</code>	card (heavier)	amethyst top strip + shadow	rule, heuristic
<code>.box-warning</code>	<code>.card-warning</code>	card (heavier)	red top strip + shadow	methodological warning

.box-question — interaction, MCQ

QUESTION FOR REFLECTION

What percentage of penguin body-mass variability is explained by bill length?

A) less than 30 % **B)** 30–60 % **C)** more than 60 %

```
1 ::: {.box-question}
2 [Question for reflection]{.panel-title}
3
4 Question text + choices A / B / C.
5 :::
```

= .panel-question · orange left
border · heading {.panel-title}

.box-result — main result

WHAT WE SEE

Each millimetre of bill length corresponds on average to **+8.7 g** body mass (95% CI: 6.2–11.2 g). The model explains 45 % of body-mass variability.

```
1 ::: {.box-result}
2 [What we see]{.panel-title}
3
4 Main conclusion or analysis result.
5 :::
```

= .panel-result · indigo left border
· heading {.panel-title} or
{.panel-lead}

.box-data — dataset description

PALMER PENGUINS

344 penguins from three islands · 3 species

(*Adelie*, *Chinstrap*, *Gentoo*)

Variables: bill length and depth (mm), flipper length (mm), body mass (g), sex, year

```
1 ::: {.box-data}
2 [Dataset name]{.panel-title}
3
4 Brief description: number of
5 observations, variables, source.
6 :::
```

= .panel-data · olive left border ·
heading {.panel-title}

.box-warning — methodological warning

CAUTION — CORRELATION ≠ CAUSATION

A longer bill does not imply that *the bill causes* greater body mass — both traits may reflect overall body size.

```
1 ::: {.box-warning}
2 [Warning heading]{.card-title}
3
4 Methodological error, misinterpretation,
5 data limitation.
6 :::
```

= .card-warning · **card** · red top
strip + shadow · heading {**.card-**
title}

.box-rule — rule, heuristic

MEAN OR MEDIAN?

If the data are symmetric → mean.

If the data are skewed or contain outliers → median.

```
1 ::: {.box-rule}
2 [Rule heading]{.card-title}
3
4 Condition → recommendation.
5 Or a verbal heuristic.
6 :::
```

= .card-rule · **card** · amethyst top
strip + shadow · heading { .card-
title }

Combination: `.box-question` + `.takeaway-strip`

QUESTION FOR REFLECTION

Why does bill length differ between penguin species and within a single species?

Morphology as adaptation. Bill length reflects life strategy — Gentoo penguins forage differently from Adelie.

```
1 ::: {.box-question}
2 [Question]{.panel-title}
3 Question...
4 :::
5
6 ::: {.takeaway-strip}
7 **Conclusion.** Closing the conclusion.
8 :::
```

`.takeaway-strip` after `.box-question` closes a slide sequence.
One slide = one question + one conclusion.

Fragment: incremental reveal of `.box-result`

Let's look at the linear model output.

RESULT

Bill length coefficient: **+8.7 g/mm** (95% CI: 6.2–11.2 g). The effect is statistically and biologically significant.

```
1 Sentence before the pause.  
2  
3 ::: {.fragment}  
4 ::: {.box-result}  
5 [Result]{.panel-title}  
6 Text that is revealed.  
7 :::  
8 :::
```

`::: {.fragment}` wraps any `.box-*` block. Students see the sentence → click → result.

Tool: plain question without a frame

How does the confidence interval change if we double the sample size?

A) Narrows B) Stays the same C) Widens

Plain slide: bold question + choices without a frame. Maximum visual contrast against slides with `.box-*` components.

Roughnotation — use sparingly

Each millimetre of penguin bill length corresponds on average to +8.7 g body mass — the effect is biologically significant.

```
1 [text]{.rn-underline-indigo}  
2 [text]{.rn-box-orange}  
3 [text]{.rn-circle-amethyst}  
4 [text]{.rn-bracket-indigo}
```

```
1 [text]{.rn-highlight-orange}  
2 [text]{.rn-cross-graphite}
```

Colours: indigo, orange,
amethyst, olive, graphite, red

Sparingly: max 1–2 effects per slide, 3–5 per lecture. Types: underline, box, circle, bracket, highlight, cross. Not as decoration — only for one key moment in a concept.

Reference Slides: SCSS Components

Cards

Card: basic structure

CARD HEADING

Card body — one or two short paragraphs. The font is intentionally smaller than the base slide size (0.82 em) to fit more content.

- Cards can also contain a list
- Each item on its own line

Syntax: `::: {.card-indigo} + [text]{.card-title}` for the coloured top strip — without the old `.slide-card` + variant combination.

Card: .card-title

THIS IS .CARD-TITLE

`[text]{.card-title}` must be the **first thing** inside the card — it creates the coloured top strip.

```
1 ::: {.card-insight}
2 [Section name]{.card-title}
3
4 Card body.
5 :::
```

Card: compact variant

STANDARD

`.card-orange` — padding `0.7rem 1rem`.
Suitable for longer content.

COMPACT

`.card-orange .card-compact` —
reduced padding (`0.45rem 0.75rem`).
For denser layouts.

Card: overview of variants

INDIGO <code>.card-indigo</code>	ORANGE <code>.card-orange</code>	AMETHYST <code>.card-amethyst</code>	OLIVE <code>.card-olive</code>	RED <code>.card-red</code>	
QUESTION <code>.card-question</code>	INSIGHT <code>.card-insight</code>	WARNING <code>.card-warning</code>	DATA <code>.card-data</code>	RESULT <code>.card-result</code>	RULE <code>.card-rule</code>

Semantic variants are aliases: `.card-question` = orange, `.card-insight` = `.card-result` = indigo, `.card-warning` = red, `.card-data` = olive, `.card-rule` = amethyst. Shorthand: `.box-question`, `.box-result`, `.box-data`, `.box-warning`, `.box-rule`.

Panels

Panel: `.panel-result`

KEY INSIGHT

Purple left border. For the main message of a block — conclusion, analysis result, or key number.

```
1 ::: {.panel-result}
2 [Heading]{.panel-title}
3 Panel text.
4 :::
```

`.panel-title` is `display: block;`
gives the heading h3 appearance
without `###`.

Panel: `.panel-question`

CONCLUSION OR OPTION

Orange left border. For conclusions, comparison results, or offering multiple answer choices.

```
1 ::: {.panel-question}
2 [Conclusion or option]{.panel-title}
3 Panel text.
4 :::
```

Panel: `.panel-question` + `.panel-lead`

Computation result

Large highlighted result — replaces the old `.answer-title`. Use at most once per slide.

```
1 ::: {.panel-question}
2 [Computation result]{.panel-lead}
3
4 Text below the result.
5 :::
```

`.panel-lead` is large and bold.
Suitable for a key number or conclusion.

Panel: `.panel-note` and `.panel-warning`

INSTRUCTIONS OR OUTPUT

Neutral olive border. For instructions, workflow steps, or model output.

CAUTION!

Red left border. For warnings, common errors, or deliberate alerts.

Panel: overview of colour variants

INDIGO

```
.panel-  
indigo
```

ORANGE

```
.panel-  
orange
```

AMETHYST

```
.panel-  
amethyst
```

OLIVE

```
.panel-  
olive
```

RED

```
.panel-red
```

Semantic variants are aliases: `.panel-result` = indigo, `.panel-question` = orange, `.panel-warning` = red, `.panel-note` = amethyst, `.panel-data` = olive. Shorthand (`.box-*`) wraps the matching panel.

Takeaway strip

Takeaway strip: standard

Sentence that leads into the conclusion — contextual paragraph above the strip.

Main slide conclusion. `.takeaway-strip` — indigo-to-orange gradient background, indigo left border. One per slide, placed at the bottom.

::: `{.takeaway-strip}` — the gradient is built in, no extra class needed.

Takeaway strip: compact variant

Sentence above the strip as usual.

Main conclusion — standard strip.

Supplementary conclusion — `.strip-compact` reduces `margin-top`. Useful for two conclusions together.

Strip: colour variants

`.strip-`
`indigo`

`.strip-`
`orange`

`.strip-`
`amethyst`

`.strip-`
`olive`

`.strip-red`

Blockquote

Blockquote div: `.blockquote`

Definition or quotation. `.blockquote` — light purple background, indigo left border (6px), rounded corners. Font 0.82 em.

- Can contain a list
- Each item on its own line

Use `::: {.blockquote}`, not Markdown `>`. The div variant gets correct styling; Markdown `blockquote` gets different styles from `presentation_theme.scss`.

Slide backgrounds

Backgrounds: dark variants

This slide has the class `{.bg-indigo}`. Text and headings are automatically white.

Syntax — add the class to the slide heading:

```
1 ## Slide title {.bg-indigo}
```

Available dark variants:

- `{.bg-indigo}` — primary indigo
- `{.bg-amethyst}` — lighter purple
- `{.bg-olive}` — grey-green
- `{.bg-red}` — red
- `{.bg-graphite}` — dark grey

Backgrounds: light variants

This slide has the class `{.bg-parchment}`. Text is automatically graphite.

Available light variants:

- `{.bg-orange}` — orange
- `{.bg-parchment}` — cream
- `{.bg-white}` — pure white

Text and heading colour adapts automatically — no need to add `.text-color-*`.

Inline highlights

Inline highlight: `.text-hgl-{color}`

The model intercept is -153.4 and the length coefficient is `4.05 g/mm` — each millimetre of length corresponds on average to four grams of body mass.

Caution — this procedure works `only when the normality assumption is met`.

Syntax: `[text]{.text-hgl-indigo_velvet}` — coloured chip with contrasting text.

Available colours: `indigo_velvet`, `orange`, `amethyst`, `grey_olive`, `graphite`, `white`, `light_gray`, `parchment`.

Inline highlight: `.text-bg-{color}` and `.text-color-{color}`

`.text-hgl-indigo_velvet` — chip with contrasting text

`.text-bg-indigo_velvet` — chip without text colour change

`.text-color-indigo_velvet` — text colour only

`.text-color-indigo_velvet`
`.text-bold` — colour + bold

```
1 [text]{.text-hgl-indigo_velvet}
2 [text]{.text-bg-indigo_velvet}
3 [text]{.text-color-indigo_velvet}
4 [text]{.text-color-indigo_velvet .text-bold}
```

Roughnotation

Roughnotation: shorthand syntax

Shorthand `{.rn-{type}-{colour}}` expands via `rn-shorthand.lua`:

Short form (new):

Each millimetre of length corresponds to approximately four grams of body mass.

```
1 [text]{.rn-underline-indigo}
```

Long form (still functional):

```
1 [text]{.rn-fragment rn-type=underline
2   rn-color="#5D2890"}
```

Available types: underline, box, circle, highlight, cross, bracket. Colours: indigo, orange, amethyst, olive, graphite, red.

Roughnotation: box, circle, bracket

box

$$R^2 = 0.76$$

```
1 [0.76]{.rn-box-indigo}
```

circle

The relationship is
strong.

```
1 [strong]{.rn-circle-orange}
```

bracket

$$\hat{y} = \beta_0 + \beta_1 x$$

```
1 [\beta_1 x]{.rn-bracket-indigo}
```

Animation fires on fragment arrival. Do not use both colours on the same slide.

Fragments and progressive disclosure

Fragment: `.fragment` block

A sentence that is on the slide from the start.

CONCLUSION ARRIVES AFTER PAUSE

This entire block appears as a single fragment — the student sees the sentence above and only after clicking does the panel with the conclusion reveal itself.

Syntax: `::: {.fragment}` wraps any block. Works with panels, blockquotes, strips, and cards.

Fragment: `.incremental` list

Which properties must a good correlation coefficient satisfy?

- It must be symmetric: correlation of X with Y = correlation of Y with X.
- It must be dimensionless: independent of the scale of variables.
- It must be in the interval -1 to $+1$.
- $+1$ = perfect positive dependence, -1 = perfect negative dependence.

Each item appears on click.

Typography

Typography: sizes

`.text-size-heading` — 140 % of base

`.text-size-large` — 130 % of base

Normal text (100 %, base 30 px)

`.text-size-smaller` — 82 % of base

`.text-size-tiny` — 65 % of base

```
1 [text]{.text-size-heading}
2 [text]{.text-size-large}
3 [text]{.text-size-smaller}
4 [text]{.text-size-tiny}
```

`.text-size-body` = 100 % (explicit reset). Old `.text-larger` / `.text-smaller` / `.text-tiny` are removed.

Typography: weight, alignment, font

`.text-center` — horizontal centre

`.text-right` — right alignment

`.text-font-heading` — Source Sans 3, heading weight

`.text-font-monospace` — JetBrains Mono

`.text-bold` — **bold**

Colour utilities

Colours: indigo_velvet and amethyst

`.bg-indigo_velvet` — primary
indigo

`.text-color-indigo_velvet`

`.text-hgl-indigo_velvet`

`.bg-amethyst` — lighter purple

`.text-color-amethyst`

`.text-hgl-amethyst`

Colours: orange and graphite

`.bg-orange` — vivid orange

`.text-color-orange`

`.text-hgl-orange`

`.bg-graphite` — dark base

`.text-color-graphite`

`.text-hgl-graphite`

Colours: neutral palette tones

`.bg-parchment`

`[.text-color-
parchment]`

`.text-hgl-
parchment`

`.bg-grey_olive`

`.text-color-
grey_olive`

`[.text-hgl-
grey_olive]`

`.bg-light_gray`

`[.text-color-
light_gray]`

`.text-hgl-
light_gray`

`.bg-white`

`[.text-color-
white]`

`.text-hgl-white`

Layout utilities

Layout: `.columns.align-top`

Left column starts at the top — without `.align-top` the content would be vertically centred.

Card in the left column.
Second paragraph inside the card — the left column is taller.

Right column is shorter.

Without `.align-top` the right text would shift down to the centre of the column pair.

Syntax: `::: {.columns .align-top}` — add the class `.align-top` to `.columns`.

Layout: vertical offsets

Example: `.slide-margin-top-15`

This card would have `margin-top: 15%` if it had the class `.slide-margin-top-15`.

Useful for vertically centring an isolated element on a slide.

`.slide-margin-top-15` → `margin-top: 15%` · `.slide-margin-top-25` → `margin-top: 25%` · `.center-vertical` → absolute centre

For a group of related elements, prefer the normal-flow `.layout-stack.layout-center-y` utility shown below.

Layout: `.layout-grid-2`

LEFT RESULT

Two equal-width elements form one comparison.

RIGHT QUESTION

The shared gap keeps the pair readable.

Use `::: {.layout-grid-2}` for a comparison or prompt pair—not merely to decorate an ordinary list.

Layout: vertical stack

First related teaching element.

Second element with consistent breathing room.

The default gap is `1.4rem`; override `--layout-gap` on the wrapper only when the rendered composition needs it.

Layout: centred stack

Which observation would change your conclusion?

`.layout-center-y` centres the short stack in normal flow without transforming its children.